Graded Assessment - Chinese Remainder Theorem

Review Learning Objectives

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Due Apr 14, 11:59 PM IST **Attempts** 3 every 8 hours

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1. If the CRT moduli are (5,7,8,9), what is the overall modulus? **1 / 1 point**

- (●) 2,520
- ( ) 9
- ( ) 29
- ( ) 210

Correct
The overall modulus is the product of the CRT moduli.

2. What is the CRT representation of 12345 mod (5,7,8,9)? **1 / 1 point**

- ( ) (6,4,1,0)
- (●) (0,4,1,6)
- ( ) (0,1,4,6)
- ( ) (6,1,4,0)

Correct
Reduce the number by each modulus in turn.

3. A integer has a CRT representation of (0,4,0,3) mod (5,7,8,9). What do we know about this number? **1 / 1 point**

- ( ) It is divisible by 5 and 8, but not necessarily 40.
- ( ) It is divisible on by 40 (or multiples thereof).
- ( ) It is divisible only by 5 and 8 (and any multiples of either).
- (●) It is divisible by both 5 and 8, but may be divisible by other prime factors as well.

Correct
Divisibility by 5 and 8 is obvious, but it is also divisible by 3, as indicated by the residue of 3 (mod 9), which is less obvious.

4. Which of the following are suitable Chinese Remainder Theorem moduli for an overall modulus of 5,040 **1 / 1 point**

- (●) (9,16,35)
- ( ) (2,3,5,7)
- ( ) (5,7,8,18)
- ( ) (2,3,4,5,6,7)

Correct
The moduli are pairwise co-prime and the produce is the overall modulus.

5. With CRT moduli of (2,3,5), what is the coefficient of the mod-3 residue when converting back to an integer? **1 / 1 point**

- ( ) 1
- (●) 10
- ( ) 3
- ( ) 30

Correct
This is congruent to 1 (mod 3), and 0 (mod 2) and (0 mod 5)

6. What is the least integer residue of the CRT representation is (4,4,4) mod (7,8,9)? **1 / 1 point**

- ( ) 503
- ( ) 17
- ( ) 288
- (●) 4

Correct
If all of the residues are equal and smaller than the smallest moduli, then they are equal to the least integer residue.

7. Using CRT moduli (7,8,9), what is the sum of 12345 and 82734? **1 / 1 point**

- (●) (5,7,3)
- ( ) (1,6,6)
- ( ) (4,1,6)
- ( ) (6,7,8)

Correct
The residues of the sum are the sums of the residues.

8. Using CRT moduli (7,8,9), what is the product of 12345 and 82734? **1 / 1 point**

- ( ) (6,7,8)
- ( ) (5,7,3)
- ( ) (6,2,8)
- (●) (4,6,0)

Correct
The residues of the product are the products of the residues.

9. Using CRT moduli (7,8,9), what is the multiplicative inverse of 46189? **1 / 1 point**

- (●) (5,5,1)
- ( ) (-3,-5,-1)
- ( ) (1,1,1)
- ( ) The multiplicative inverse does not exist.

Correct
The residues of the multiplicative inverse are the multiplicative inverses of the residues.

10. Using CRT moduli (7,8,9), what is 12345 to the 5th power? **1 / 1 point**

- (●) (2,1,0)
- ( ) (4,6,2)
- ( ) (6,7,8)
- ( ) (0,0,0)

Correct
The residues of a power are the powers of the residue.